QQ-T-00276a (DSA-DM)
March 19, 1970
Interim Revision of
Fed. Spec. QQ-T-276
July 18, 1951

INTERIM FEDERAL SPECIFICATION

# TIN FOIL, DENTAL

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification QQ-T-276.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers tin foil suitable for dental laboratory use.

1.2 Classification.

1.2.1 Sizes. The tin foil shall be of the following sizes, as specified (see 6.1):

Size 1. One-pound roll.
Size 2. Half-pound roll.

1.2.2 Thicknesses. The tin foil shall be of the following thicknesses, as specified (see 6.1):

Thickness A. 0.0010 inch.
Thickness B. 0.0020 inch.
Thickness C. 0.0030 inch.

FSC 9535

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2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| MMM-A-250 | Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes). |
| PPP-B-566 | Boxes, Folding, Paperboard. |
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Box, Fiberboard. |
| PPP-B-676 | Box, Setup. |
| PPP-T-76 | Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing). |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-123 | Marking for Domestic Shipment (Civilian Agencies). |

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston; New York; Atlanta; Chicago; Kansas City, Mo.; Fort Worth; Denver; San Francisco; Auburn, Washington; and Washington, D. C.



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STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

(Copies of Military specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Foil. The tin foil shall contain not less than 99.80 percent tin.

3.1.2 Spool. The spool on which the foil is wound shall be fabricated from cardboard, plastic, or other suitable material.

3.2 Style, design, and dimensions. The tin foil shall be suitable for dental laboratory use in the construction of dentures. The foil shall serve as a burnished covering, so that a smooth, accurate surface may be imparted to the finished product. The foil shall be 6 ± 1/32 inches wide. The specified thicknesses of 0.0010, 0.0020, and 0.0030 inch shall each have a plus tolerance of 0.0003 inch, but a negative tolerance shall not be permitted. The spool shall be of standard commercial design. As an alternate design, the spool shall have sidewalls to protect the edges of the roll of foil.

3.3 Sizes furnished. The tin foil shall be furnished as a single, continuous roll containing a minimum of one-half pound or one pound of foil, as applicable. Each roll shall be evenly wound on a spool of appropriate size.



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3.4 Finish. The tin foil shall have a smooth, bright polished finish.

3.5 Workmanship. Workmanship shall be first class throughout. The tin foil shall be free from holes, breaks, folds, or wrinkles and other defects which may detract from its appearance or impair its serviceability. The foil shall be furnished clean, and free from dirt, oil, grease, wax, or other particulate matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the material, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in Tables I and III (see 4.2.1 and 4.3) and Table II (see 4.2.2).



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4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in Table I. Unit of product for examination purposes shall be one roll of foil.

Table I. Examination criteria

| Type of defect | Inspection level | AQL (percent defective) |
|---|---|---|
| Major A | S-2 | 1.0 |
| Major B | S-2 | 1.5 |
| Minor A | II | 2.5 |
| Minor B | S-2 | 4.0 |

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one tin foil roll.

Table II. Test criteria

| Characteristic | Requirement | Test Method | Inspection level |
|---|---|---|---|
| Chemical composition | 3.1 | 4.4.1 | S-1 |

4.3 Examination. The tin foil rolls shall be examined for defects including, but not limited to, those listed in Table III.

Table III. Classification of defects

| Category | Defect |
|---|---|
| Major A: | |
| 101 | Tin foil not free from breaks, tears, holes, folds, or wrinkles. |
| 102 | Tin foil not furnished clean and free from dirt, oil, grease, wax, or other particulate matter. |
| 103 | Style and design not as specified. |
| 104 | Foil not furnished as a single, continuous roll. |



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Table III. Classification of defects (Con't)

| | |
|---|---|
| **Major B:** | |
| 151 | Weight of foil on spool not as specified. |
| 152 | Thickness of foil not as specified. |
| **Minor A:** | |
| 201 | Finish not bright and smooth. |
| 202 | Individual roll not evenly wound on spool of appropriate size. |
| 203 | Packaging and packing not as specified. |
| 204 | Marking on packaging and pack not as specified. |
| **Minor B:** | |
| 251 | Width of foil not as specified. |

4.4 Test procedures. Chemical composition.

4.4.1 Percentage of tin. Specimens shall be taken from each of the rolls and the following analysis shall be made. Weigh accurately about 0.5 gram of tin foil and dissolve in 25 milliliters of concentrated HCl by heating in a 500-milliliter covered beaker. Add bromine water until a permanent yellow color is obtained. Dilute the solution to 100 milliliters with cold water, add methyl red indicator, and neutralize the solution with ammonia. Add about 50 milliliters of saturated solution of ammonium nitrate and dilute the solution to 300 milliliters with boiling water, stirring the contents constantly. Allow the beaker to stand on the steam bath until the precipitate of $SnO_2$ has settled. Decant the supernatant liquor through a hard filter paper and wash the precipitate six times by decantation, using 100 milliliters of boiling one-percent $HNO_3$ and allow the precipitate to settle after each washing. After the precipitate has been completely transferred to the filter paper, thoroughly wash and allow to drain; transfer to a porcelain crucible; dry on a hot plate; ignite at a low temperature until the filter paper has been consumed; and finally dry to constant weight. Calculate the tin content as follows:

$$\frac{\text{Weight of } SnO_2 \times 0.78766 \times 100}{\text{Weight of sample in grams}} = \text{Percentage of tin}$$



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## 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packing shall be level A, or C, as specified (see 6.1).

5.1.1 Unit of issue. Each 1/2 pound roll of tin foil, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging quantities

| Unit package | Intermediate package | Exterior container |
|---|---|---|
| 1 unit | 6 units | 96 units |

5.1.2.1 Variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit of foil shall be evenly wound on a spool of appropriate size. Each unit shall be wrapped in tissue paper and packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors are acceptable. Closure of box shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.



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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B,or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special markings required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military requirements.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.



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5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and thickness required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with an asterisk to indicate where changes from the previous issue have been made.

6.3 Limitations. This specification does not cover all types, classes, grades, styles, thicknesses, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
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Preparing activity:
Defense Supply Agency - DM

Review activities:
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Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.



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**CAUTION**

**NOTICE TO BIDDERS/OFFERORS**

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

**INSTRUCTIONS:** This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

**SPECIFICATION** QQ-T-00276a(DSA-DM) FSN:

**ORGANIZATION**

**CITY AND STATE** | **CONTRACT NUMBER**

**MATERIAL PROCURED UNDER A**

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

**1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?**

**A. GIVE PARAGRAPH NUMBER AND WORDING.**

**B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES**

**2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID**

**3. IS THE SPECIFICATION RESTRICTIVE?**

☐ YES ☐ NO *(If "yes", in what way?)*

**4. REMARKS** *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

**SUBMITTED BY** *(Printed or typed name and activity - Optional)* | **DATE**

**DD FORM 1 JAN 66 1426** REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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